# Ronald David Martinez Marenco

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## Summary

Software Engineer and Computer Science Student-Researcher with industry experience building AI-driven solutions and web applications using TypeScript, Python, Next.js, and FastAPI. Combining strong academic foundations with production-ready software engineering.

## Work Experience

**Full-Stack Software Engineer** at Redwood Tech, Remote (Contract) *May 2026 - Present*

- Engineered end-to-end full-stack features using TypeScript, Python, React, Next.js, and FastAPI across scalable web applications, backend services, and external API integrations.
- Implemented and evolved conversational AI agents equipped with tool calling capabilities, contextual memory, speech-to-text transcription, and multimodal content processing.
- Built operational automation workflows and human-in-the-loop mechanisms to manage agent escalation, fallback logic, and conversational state retention.
- Developed custom Excel-based data loaders and administrative web interfaces to streamline structured data ingestion and internal business operations.
- Extended user permission architectures, change-auditing capabilities, and relational data persistence using PostgreSQL, Redis, Prisma, and Docker.

**Frontend Developer** at Fundación Remedios de Manasés, Remote (Freelance) *November 2024*

- Developed and deployed a responsive web platform using React and JavaScript, aligning layout, typography, and color schemes with institutional branding guidelines.
- Designed custom CSS architecture and breakpoints to ensure optimal accessibility, visual clarity, and cross-device performance across desktop and mobile browsers.
- Implemented interactive UI components and lightweight CSS transition effects to optimize page load speeds and user engagement.

## Projects

**Data Viento: Weather Climate Insights Platform -** DataViento Webpage *October 2025 - January 2026*

Python, FastAPI, JavaScript, HTML, CSS, MySQL, Docker, OpenMeteo API, Cron Automation, Gemini LLM

- Developed a full-stack web application that collects, stores, and visualizes climate data from the OpenMeteo API, providing interactive dashboards and AI-assisted explanations based on the user's selected city.
- Built backend services using FastAPI to consume weather data from the OpenMeteo API and store structures time-series data in MySQL, ensuring efficient retrieval and analysis.
- Implemented frontend interface with HTML, CSS,JavaScript and chart.js library to render a dashboard that displays weather variables and trend charts based on the user's selected city.
- Developed an Gemini LLM-powered chatbot to explain weather variables, chart interpretations, and climate impacts applying prompt engineering to improve contextual responses.
- Automated data ingestion via Python Scripts and Linux cron jobs, updating database on a schedule without manual intervention.
- Docker environment for containerized deployment and Git version control to manage the full project lifecycle collaboratively.

**Hunting Exoplanets NASA-Hackathon** - Hunting Exoplanet Webpage *October 2025*

Python, Tensorflow, Google Colab, Streamlit.

- Developed a predictive web application for exoplanet classification, allowing users to input planetary features or upload CSV files for batch prediction using an interactive Streamlit interface.
- Cleaned and preprocessed astronomical datasets in Google Colab, handling missing values using median imputation and performing correlation analysis to understand feature relationships.
- Built and evaluated multiple ensemble machine learning models (Voting Classifier combining RandomForest, LGBM, XGBoost) within a TensorFlow-compatible architecture to ensure robust prediction performance.
- Achieved approximately 87% accuracy on the trained dataset.

**Educational AI Chatbot with RAG and Voice Interface** - Github Repository *April 2025 - June 2025*

Java, Spring Boot, Spring AI, MySQL, HTML, Tailwind CSS, JavaScript, Gemini (Vertex AI), ElevenLabs, Jina AI Embeddings

- Developed an educational chatbot backend using Spring Boot and Spring AI, supporting conversational memory and intent-aware responses.
- Integrated generative responses via Gemini (Vertex AI) and added text-to-speech (TTS) functionality using ElevenLabs for voice interaction.
- Implemented Retrieval-Augmented Generation (RAG) by extracting text from educational documents using Apache Tika and generating embeddings with Jina AI, enabling document-based question answering.
- Built a lightweight frontend with Tailwind CSS and persisted conversation and metadata in a MySQL database.

**Sign Language Recognition App Using Machine Learning** - Scientific article *March 2024 - Present*

Python, Tensorflow, Keras, Mediapipe Holistic, OpenCV, Numpy, LSTM / CNN Neural Networks

- Developed a sign language recognition application using computer vision and deep learning to classify 10 Colombian sign gestures.
- Created a custom dataset by recording and labeling sign language samples, then expanded it using image and sequence augmentation techniques.
- Extracted pose, hand, and facial landmarks with MediaPipe Holistic and standardized input data before model training.
- Trained an LSTM-based neural network using TensorFlow and Keras to model temporal gesture sequences.
- Achieved approximately 95% classification accuracy on the trained gesture set through iterative training and evaluation.

## Skills

- **Languages:** Python, TypeScript, Java, SQL.
- **Frontend & Web:** React, Next.js, HTML5, CSS3, Tailwind CSS.
- **Backend & APIs:** FastAPI, Spring Boot, Prisma (ORM), RESTful APIs.
- **AI & Machine Learning:** AI Agents, RAG, TensorFlow, Keras, OpenCV, YOLO, MediaPipe, Pandas, NumPy.
- **Databases & Infrastructure:** PostgreSQL, MySQL, Redis, Docker, Git, GitHub, Linux / WSL2.

## Education

**B.Sc. in Systems Engineering** from Universidad Simon Bolivar, Colombia *In Progress - 6th Semester*

- 1st Place in GPA - Systems Engineering Program
  GPA: 4.85 / 5.0
- Active member of the Data Science research group.
- Student representative for the systems engineering program.

## Honor & Awards

- **NASA Space Apps Challenge - Global Nominee**
  **Project**: Hunting Exoplanets - Developed an AI-based solution for detection of exoplanets using Python, Tensorflow, Pandas, Streamlit.
- **Elite Researcher Recognition**
  Awarded for academic research excellence.
- **RedColsi Speaker at international conference**
  Speaker for **Sign Language Recognition App Using Machine Learning**